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Engineering Mathematics

3Week [18 September - 24 September]

Quiz 3

Started on	2023-09-25 01:03
State	Finished
Completed on	2023-09-25 01:31
Time taken	27 mins 17 secs
Grade	6.00 out of 10.00 (60%)

Question 1

Incorrect

Mark 0.00 out of 1.00

Which of the equations below admits a special integrating factor?

Select one:

- ☐ $(2x^2 + y)dx + (x^2y - x)dy = 0$
- ☐ $(y^2 + xy)dx - x^2dy = 0$
- ☐ $(x^2 \sin x + 4y)dx + xydy = 0$
- ☒ $(2y^2x - y)dx + 2xdy = 0$ ✖

Your answer is incorrect.

The correct answer is: $(2x^2 + y)dx + (x^2y - x)dy = 0$

Question 2

Correct

Mark 1.00 out of 1.00

Find an integrating factor of the equation

$(2y^2 - 6xy)dx + (3xy - 4x^2)dy = 0$ in the form

$x^n y^m$.

Select one:

- ☒ xy ✔
- ☐ x^2y
- ☐ xy^2
- ☐ x^2y^2

Your answer is correct.

The correct answer is: xy

Question 3

Incorrect

Mark 0.00 out of 1.00

Among the equations below choose a homogeneous differential equation.

Select one:

- ☐ $(x - 2y)dx + (2x - y)dy = 0$
- ☐ $(x - 2y + 1)dx + (y + x)dy = 0$
- ☐ $(x^3 + y^3)dx + (\cos x + \sin y)dy = 0$
- ☒ $(x^2 - 3y^3)dx + 2xydy = 0$ ✗

Your answer is incorrect.

The correct answer is: $(x - 2y)dx + (2x - y)dy = 0$

Question 4

Correct

Mark 1.00 out of 1.00

Choose a correct substitution for the following homogeneous differential equation $(t^2 - tx + x^2)dt + 2txdx = 0$.

Select one:

- ☒ $v = \frac{x}{t}$ ✓
- ☐ $v = \frac{y}{x}$
- ☐ $v = xt$
- ☐ $v = x - t$

Your answer is correct.

The correct answer is: $v = \frac{x}{t}$

Question 5

Incorrect

Mark 0.00 out of 1.00

Among the equations below choose a differential equation NOT in the form $\frac{dy}{dx} = G(ax + by)$.

Select one:

- ☐ $\frac{dy}{dx} = \frac{2x-y}{2y-x}$
- ☒ $\frac{dy}{dx} = \sqrt{x+y} - 1$ ✗
- ☐ $\frac{dy}{dx} = \frac{x-y-1}{2y-2x}$
- ☐ $\frac{dy}{dx} = x^2 + 2xy + y^2$

Your answer is incorrect.

The correct answer is: $\frac{dy}{dx} = \frac{2x-y}{2y-x}$

Question 6

Correct

Mark 1.00 out of 1.00

Choose a correct substitution for the following differential equation

$$\frac{dy}{dx} = \sin(x - y)$$

Select one:

☒ $v = x - y$ ✓

☐ $v = \frac{y}{x}$

☐ $v = \sin(x - y)$

☐ $v = \sin x \cos y$

Your answer is correct.

The correct answer is: $v = x - y$

Question 7

Correct

Mark 1.00 out of 1.00

Among the equations below choose the Bernoulli differential equation.

Select one:

☒ $\frac{dy}{dx} = 5y - xy^3$ ✓

☐ $\frac{dy}{dx} = \sqrt{x + y} - 1$

☐ $\frac{dy}{dx} = \frac{x - y - 1}{2y - 2x}$

☐ $\frac{dy}{dx} = x^2 + 2xy + y^2$

Your answer is correct.

The correct answer is: $\frac{dy}{dx} = 5y - xy^3$

Question 8

Incorrect

Mark 0.00 out of 1.00

Choose a correct substitution for the following differential equation

$$\frac{dy}{dx} = \frac{y}{x-2} + 5(x-2)y^{1/2}$$

Select one:

☐ $v = y^{1/2}$

☒ $v = \frac{y}{x-2}$ ✗

☐ $v = x - 2$

☐ $v = y^{-1/2}$

Your answer is incorrect.

The correct answer is: $v = y^{1/2}$

Question 9

Correct

Mark 1.00 out of 1.00

Among the equations below choose the differential equation with linear coefficients.

Select one:

☒ $\frac{dy}{dx} = \frac{2x-3y+7}{5y-x-3}$ ✓

☐ $\frac{dy}{dx} = \sqrt{x+y} - 1$

☐ $\frac{dy}{dx} = \frac{x^2+y^2}{3xy}$

☐ $\frac{dy}{dx} = 5(x-2)y^2 - (2x+1)y$

Your answer is correct.

The correct answer is: $\frac{dy}{dx} = \frac{2x-3y+7}{5y-x-3}$

Question 10

Correct

Mark 1.00 out of 1.00

Choose a correct substitution for the following differential equation

$$\frac{dy}{dx} = \frac{3x-y-6}{x+y+2}$$

Select one:

☒ $u = x - 1, \quad v = y + 3$ ✓

☐ $u = x - 6, \quad v = y + 2$

☐ $u = x - 3, \quad v = y + 1$

☐ $u = x - 2, \quad v = y + 6$

Your answer is correct.

The correct answer is: $u = x - 1, \quad v = y + 3$